

Class 4: Homogeneous and Nonhomogeneous Systems

A homogeneous system is one of the form

Claim: Every homogeneous system is consistent. Why?

Big question for a homogeneous system is whether the solution is unique or infinite. How do you decide?

Examples: investigate the solution sets for the homogeneous systems with the given matrices.

$$1. \quad A = \begin{bmatrix} 0 & 7 & 9 \\ 2 & 1 & -1 \\ 5 & 6 & 2 \end{bmatrix}$$

$$2. A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 8 & 9 \\ 2 & -1 & 2 \end{bmatrix}$$

$$3. A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A nonhomogeneous system of equations has the form

A nonhomogeneous system may or may not be consistent.

Key idea: The solution set for a consistent nonhomogeneous system is the same type as the solution set of the corresponding homogeneous system.

Example 4: As we saw in our last class, the system $A\mathbf{x} = \mathbf{b}$ is solved by row reducing the augmented matrix.

Suppose we get $[A : \mathbf{b}] \sim \begin{bmatrix} 1 & 2 & 3 & : & 4 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$

Theorem 6: If $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{p}$ is a solution to the system, then every vector in the solution set has the form $\mathbf{p} + \mathbf{v}$ where $\mathbf{v}$ is a solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$.